

QI WANG

CONTACT INFORMATION

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ACADEMIC APPOINTMENT

Postdoctoral Fellow in Biostatistics, Emory University

August 2026–Present

Department of Biostatistics and Bioinformatics, Rollins School of Public Health

Mentor: [Emily N. Peterson, Ph.D.](#)

EDUCATION

Ph.D. in Statistical Science, University of California, Santa Cruz

May 2026

Advisors: [Paul A. Parker, Ph.D.](#) and [Robert B. Lund, Ph.D.](#)

M.S. in Statistics, University of California, Irvine

September 2019–May 2021

B.S. in Applied Statistics, Central University of Finance and Economics

September 2015–July 2019

RESEARCH INTERESTS

Methodology: Small area estimation, Bayesian measurement error models, spatial and spatio-temporal modeling, deep learning, count time series, and change point detection.

Applications: Public health surveillance, health disparities, Indigenous population health, climate science, and environmental data analysis.

PUBLICATIONS AND MANUSCRIPTS

- **Wang, Q.**, Parker, P. A., and Lund, R. (2025). [Spatial deep convolutional neural networks](#). *Spatial Statistics*, **66**, 100883.
- **Wang, Q.**, Parker, P. A., and Lund, R. (2025). [Hierarchical count echo state network models with application to graduate student enrollments](#). *Data Science in Science*, **4**(1), 2565243.
- Kei, Y. L.[†], **Wang, Q.**[†], Parker, P., Ribalet, F., and Hyun, S. (2026). [Change point detection for cell populations measured via flow cytometry](#). *arXiv preprint arXiv:2603.05700*.
[†]Co-first authors.
- **Wang, Q.**, Parker, P. A., and Lund, R. *A spatio-temporal study of planetary snow trends*. Resubmission invited (reject and resubmit), *Journal of the American Statistical Association*.

RESEARCH EXPERIENCE AND ONGOING PROJECTS

- **Small-Area Mortality Surveillance for American Indian and Alaska Native Populations**
 - Develop Bayesian hierarchical spatio-temporal and measurement error models for county-year mortality estimation, correcting racial misclassification in death records and undercounting in population denominators.
 - Integrate state death certificates, Tribal registry data, and U.S. Census Bureau population estimates to quantify uncertainty across Tribal citizenship, racial identification, and Tribal geography.
 - Help translate the methodology into a reproducible data science pipeline for Tribal public health surveillance while supporting Tribal data sovereignty.
- **Spatial Statistics for Climate and Environmental Applications**

- Collaborate with Paul A. Parker and Robert B. Lund on spatial and spatio-temporal statistical methodology for climate and environmental data, including planetary snow trend analysis.
- **A Unified Copula Approach to Dependent Data Modeling via Neural Networks**
 - Develop a deep learning approach for estimating marginal and joint cumulative distributions of spatial, temporal, and spatio-temporal data using a Gaussian copula.
- **Bridging Probability and Nonprobability Sampling: A Review of Methods, Theory, and Integration Strategies**
 - Review theoretical frameworks and recent advances for valid inference using nonprobability samples and integrated survey data.

TEACHING EXPERIENCE

- **Graduate Student Instructor**, UCSC STAT/CSE 266A: Graduate Data Visualization and Statistical Programming in R, Spring 2025.
- **Graduate Student Instructor**, UCSC STAT 7L: Statistical Methods for the Biological, Environmental, and Health Sciences Laboratory, Summer 2021.
- **Teaching Assistant**, UCSC STAT 7: Statistical Methods for the Biological, Environmental, and Health Sciences, Fall 2021, Winter 2021, Spring 2021, and Winter 2025.
- **Teaching Assistant**, UCSC STAT 131: Introduction to Probability Theory, Fall 2023.

INVITED AND CONTRIBUTED PRESENTATIONS

- **Joint Statistical Meetings**, Topic-Contributed Session, 2025. “Hierarchical Count Echo State Network Models with Application to Graduate Student Enrollments,” in *Hierarchical Modeling and Machine Learning for Complex Survey Data*.
- **International Symposium on Forecasting**, 2025. “Hierarchical Count Echo State Network Models with Application to Graduate Student Enrollments.”
- **Joint Statistical Meetings**, Contributed Session, 2024. “Spatial Deep Convolutional Neural Networks,” in *Recent Advances in Machine Learning-Based Methods for Spatio-Temporal Analysis*.

AWARDS AND HONORS

- UCSC Dissertation-Year Fellowship (\$33,000 plus tuition), Spring 2025.
- UCSC First Generation Graduate Group Research and Travel Award, Spring 2025.
- UCSC Graduate Dean’s Research Travel Grant, Winter 2025.
- UCSC Regents Fellowship, Spring 2022.

PEER REVIEW

- *Spatial Statistics*
- *Journal of Time Series Analysis*
- *Journal of Statistical Computation and Simulation*
- *International Journal of Public Health*
- *Renewable Energies, Environment & Power Quality Journal*

PROFESSIONAL AFFILIATIONS

- Member, American Statistical Association (ASA)
- Member, Eastern North American Region (ENAR), International Biometric Society